Special instructions

- This exam consists of **three parts**. The mark breakdown is as follows:

Part	A	B	C					
Question	1	2	3	4	5	6	7	Total
Marks	4	12	8	12	12	16	16	80

- **Unless a question indicates a numeric value is required**, numeric answers may be left as numeric expressions if desired (e.g., if the correct answer was 2, an answer of '1 + 1' would receive the same credit as an answer of '2'). If you express answers as numeric values these should be accurate to at least three decimal places.

- If you are asked to identify the type of a variate, choose from Continuous, Discrete, Categorical, Ordinal, or Complex.

- If you believe insufficient information is provided in a question to determine the requested answer, you may write 'not enough information', 'NEI', or similar. If the question indicates you should justify your answer, this justification is still required even if your answer is 'not enough information'.

- Additional blank pages are provided at the end of the exam for rough work and/or additional space for answers. If you wish any work on these pages to be graded, you must clearly indicate this on the corresponding question page.

- Good luck :)

Part A

Question 1 [4 marks]

(a) [4 marks] Questions (i)-(iv) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) Which of statements A-C is FALSE?

A Sample variance is always non-negative.

B Sample kurtosis is always non-negative.

C Relative risk is always non-negative.

D All of the above statements are true.

(ii) A 95% confidence interval is approximately the same as:

A A 10% likelihood interval.

B A 15% likelihood interval.

C A 90% likelihood interval.

D A 97.5% likelihood interval.

(iii) When testing $H_0 : \sigma^2 = \sigma_0^2$ for the $G(\mu, \sigma)$ model using a sample of size n and the test statistic $u = \frac{(n-1)S^2}{\sigma_0^2}$, how do we calculate the p-value?

A $p = 2P(U \leq u), U \sim \chi^2_{n-1}$

B $p = 2P(U \geq u), U \sim \chi^2_{n-1}$

C $p = 2P(U \leq u), U \sim \chi^2_{n-2}$

D It depends on the value of u.

(iv) Define a such that $P(Z \leq a) = 0.975, Z \sim G(0, 1)$ and b such that $P(T \leq b) = 0.975, T \sim t_2$. Which of the following statements is TRUE?

A $a < b$

B $a = b$

C $a > b$

D Cannot be determined.

Part B

Part B is based on the assignments. All variate names refer to those in the Assignment Dataset Information file. All R commands assume the assignment dataset has been stored in R in an object named `mydata`. You may further assume that the variate `subject.age.log` has been created (as a variate within `mydata`) as described in Analysis 2 of Assignment 1.

Question 2 [12 marks]

Parts (a)-(e) are based on Analysis 2 of Assignment 4. The following R commands are run on the full assignment dataset (not just one city), with the corresponding output. Note that some output has been omitted deliberately. Note that some numbers have been rounded for your convenience. You may treat all numeric values in the output as exact.

```
> mydata$subject.age.bin <- as.numeric(mydata$subject.age >= 30)
> table(mydata$subject.age.bin)

  0   1
386 564
> vehicle.year.younger <- mydata$vehicle.year[mydata$subject.age.bin == 0]
> vehicle.year.older <- mydata$vehicle.year[mydata$subject.age.bin == 1]
> t.test(vehicle.year.younger, vehicle.year.older, var.equal = T)

t = ???, df = ???, p-value = 0.8078
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
     -0.95    0.75
sample estimates:
mean of x mean of y
 2006.800       (b)
```

(a) [1 mark] How many traffic stops in this sample were of individuals aged at least 30 years old?

(b) [2 marks] What value has been replaced by `(b)` at the end of the output? Carefully justify your answer.

(c) [2 marks] Carefully describe, using words and/or mathematical notation, the null hypothesis that is being tested using the `t.test` command in this output. Ensure any relevant modelling assumptions are also clearly described.

For your convenience, the preceding R commands and output are repeated here:

```
> mydata$subject.age.bin <- as.numeric(mydata$subject.age >= 30)
> table(mydata$subject.age.bin)

  0   1
386 564
> vehicle.year.younger <- mydata$vehicle.year[mydata$subject.age.bin == 0]
> vehicle.year.older <- mydata$vehicle.year[mydata$subject.age.bin == 1]
> t.test(vehicle.year.younger, vehicle.year.older, var.equal = T)

t = ???, df = ???, p-value = 0.8078
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
     -0.95    0.75
sample estimates:
mean of x mean of y
 2006.800       (b)
```

(d) [1 mark] What probability distribution is used to calculate the p-value seen in the output?

(e) [2 marks] Briefly summarize, with justification, the conclusion of the test in the context of the study.

Parts (f)-(g) are based on Analysis 1 of Assignment 4.
Your colleague Cyril would like to use the results of Analysis 1 to give a 95% prediction interval for the value of `vehicle.year` for someone who is 400 years old.

(f) [2 marks] Would this prediction interval be wider, narrower, or the same width as a 95% confidence interval for the mean `vehicle.year` for those who are 400 years old? Briefly justify your answer.

(g) [2 marks] Later, Cyril asks for a prediction interval for someone who is 10 years old. Briefly respond to Cyril's request. Note: You are not expected to calculate the interval in your answer.

Part C

Question 3 [8 marks]

Let $y_1, y_2, \ldots, y_n$ be independent, identically distributed observations drawn from a continuous probability distribution with parameter θ. You are told that the probability density function for this distribution is:

$$g(y;\theta) = y^{-2\theta^3} \log(y) e^{3\theta^2 y} \quad y > 1 \text{ and } \theta > 0$$

Note: This is not a real probability density function, but you may assume it has all the properties necessary in order to solve the following questions, and as such may answer the following under the assumption it is a real probability density function. In particular, you may assume that the likelihood and log-likelihood functions are maximized for the value of θ at which their first derivatives equals 0.

A sample of data $y_1, y_2, \ldots, y_n$ is observed.

(a) [4 marks] Find the maximum likelihood estimate of θ, expressed as a function of the observed data. Carefully show all your steps.

Let $V_1, V_2, \ldots, V_n$ denote a random sample from a distribution with range $y > 0$ and unknown parameter $\theta > 0$. Let $\overline{V} = \frac{1}{n}\sum\limits_{i=1}^{n} V_i$. You are told that

$$W = \frac{\sum\limits_{i=1}^{n} V_i}{\theta^2/\sqrt{n}} \sim \chi^2_n$$

is an exact pivotal quantity for θ.

A sample $v_1, v_2, \ldots, v_{25}$ is observed with sample mean 2 and sample standard deviation 10. You are given the following R output:

```
> qchisq(0.05, 24)
[1] 13.84843
> qchisq(0.75, 24)
[1] 28.24115
> qchisq(0.85, 24)
[1] 31.13246
> qchisq(0.95, 24)
[1] 36.41503
```

```
> qchisq(0.05, 25)
[1] 14.61141
> qchisq(0.75, 25)
[1] 29.33885
> qchisq(0.85, 25)
[1] 32.28249
> qchisq(0.95, 25)
[1] 37.65248
```

Note: For formatting reasons, this output is arranged in two columns. The output is immediately below its corresponding command (for example, the output of `qchisq(0.05, 24)` is `[1] 13.84843`.

(b) [4 marks] Using W, construct an 80% confidence interval for θ based on these data. **Carefully justify your answer**, showing all your steps in the derivation of the confidence interval, and clearly identifying any numeric values used in your answer.

Question 4 [12 marks]

A statistics professor was interested in learning about the consumption of coffee by professors at the University of Waterloo. In particular, the professor was interested in the distribution of the number of cups of coffee drunk per day by professors, including the mean number of cups drunk per day.

To investigate, the professor emailed all 60 professors in the Department of Statistics and Actuarial Science. The email was sent on a Monday, and asked professors how many cups of coffee they had drunk that day. 25 professors responded, with the data summarized on the following page.

(a) [4 marks] Would sample error be a concern in this study? Carefully justify your answer. If relevant to your answer, the target population, study population, and/or sampling protocol should be described.

(b) [1 mark] Would measurement error be a concern in this study? Briefly explain why or why not.

The results of the survey were as follows:

Number of Cups	0	1	2	3	4	5+
Number of Faculty	10	8	5	1	1	0

So, for example, 10 professors said they had drunk 0 cups of coffee, 8 professors said they had drunk 1 cup of coffee, and no professors said they had drunk 5 or more cups of coffee.

Let $y_1, y_2, \ldots, y_{25}$ denote the number of cups of coffee drunk by each of the professors according to the survey data. We will assume that these data represent a random sample from a Poisson(θ) distribution.

Let $Y_1, Y_2, \ldots, Y_n \sim$ Poisson(θ), independently and identically distributed, denote the number of cups consumed by each of n professors in a random sample from the study population.

(c) [2 marks] Do you have any concerns about the assumption that these data represent observations from independent and identically distributed Poisson random variables? Briefly justify your answer.

(d) [1 mark] What is $\hat{\theta}$, the maximum likelihood estimate of θ? An exact numeric value (not a numeric expression) is required for full marks.

We wish to test the null hypothesis that a Poisson model is appropriate for these data. You are given the following table of observed and expected counts under the null hypothesis:

Number of Cups	0	1	2	3	4	5+
Observed	10	8	5	1	1	0
Expected	9.197	(i)	4.598	1.533	0.383	(ii)

(e) [2 marks] Give numeric values or expressions for the values replaced by (i) and (ii). In your answers, you may use your numeric answer to part (d), or you may write $\hat{\theta}$.

(i):

(ii):

(f) [2 marks] Suppose we were to test the null hypothesis using the Pearson goodness-of-fit test. Assuming we combine categories in order to ensure all expected counts are greater than or equal to 5, what distribution would we consult to calculate the p-value?

Question 5 [12 marks]

Having built up restaurant experience at *Pita Parker*, Alex and Blake are now working on co-op at *Pizza Parker*, a superhero-themed pizza restaurant. The restaurant offers delivery, and advertises that if you do not receive your order within 30 minutes, your order is free (i.e., you do not have to pay for it). The restaurant's goal is for the delivery times of all orders in 2025 to have mean 20 minutes and standard deviation 4 minutes. The restaurant owners wish to investigate whether this goal is being met.

To avoid variation due to different order types, a random sample of 25 orders is taken from all orders for a single large cheese pizza (and no other items). Let $Y_1, Y_2, \ldots, Y_{25} \sim G(\mu, \sigma)$, independent and identically distributed, with observed values $y_1, y_2, \ldots, y_{25}$, which are measured in minutes. Suppose that these data are entered into R and stored in an object named `cheese`. You are given the following R commands and output, where some details have been deliberately omitted.

```
> mean(cheese)
[1] 18.5
> var(cheese)
[1] 9
> t.test(cheese, mu = 20)

data:  cheese
t = (b), df = ??, p-value = 0.01965
alternative hypothesis: true mean is not equal to 20
```

(a) [3 marks] What is the maximum likelihood estimate of the probability a randomly chosen delivery will take more than 30 minutes? Carefully justify your answer. You may leave your answer as a probabilistic expression. Hint: Clearly state the maximum likelihood estimates of μ and σ as part of your answer.

(b) [2 marks] Give a numeric value or expression for the value that has been replaced by `(b)`.

For your convenience, the preceding R commands and output are repeated here:

```
> mean(cheese)
[1] 18.5
> var(cheese)
[1] 9
> t.test(cheese, mu = 20)

data:  cheese
t = (b), df = ??, p-value = 0.01965
alternative hypothesis: true mean is not equal to 20
```

(c) [2 marks] Give an equation for a 92% confidence interval for μ based on these data. All terms in your equation must be clearly defined, either by numeric or probabilistic expressions. Note: If you use a quantile from a probability distribution, you <u>must</u> include a probabilistic expression to define it, you may not use a numeric value.

(d) [3 marks] Give a probabilistic expression for the p-value reported in this output. You may write 'b' for the correct answer to part (b), if you wish.

Alex decides to investigate whether only sampling orders of a single cheese pizza might have affected the results of their study. Each of the 25 customers selected in the preceding analysis placed a second order with the company a few weeks later. These second orders were all of multiple items. Alex decides to compare the delivery times of these multiple item orders with the orders that were for only one pizza. You may assume that the same customer will request both of their orders to be delievered to the same address.

Alex creates three variates in R to describe the data:

- `customer`: An id number to uniquely identify each of the 25 customers.
- `order`: The order type (single item or multiple item).
- `time`: The delivery time (in minutes).

Alex stores the data in R in a data frame called `mydata`.

```
> head(mydata)
  customer          order time
1        1    single item   21
2        1 multiple items   18
3        2    single item   17
4        2 multiple items   21
5        3    single item   24
6        3 multiple items   26
```

(e) [2 marks] Should Alex use a paired or unpaired test? Briefly justify your answer.

Question 6 [16 marks]

Tired of restaurant work, Alex and Blake have started working at *Peter's Parkas*, a store specializing in superhero-themed winter cothing.

A study is being conducted to look at the relationship between temperature and the amount of money being spent on winter clothes at the store. A random sample of days from the year 2023 was taken, and for each day two measurements are taken: the outside temperature measured at 9am (in degrees Celsius) and the total amount of money spent at the store that day (in Canadian dollars). These variates have been stored in R in objects named `temp` and `spend`, respectively.

For the analyses that follow, assume the regression model $Y_i \sim G(\alpha + \beta x_i, \sigma)$, $i = 1, 2, \ldots, n$, independently, where x_i corresponds to the 9am temperature on the i^{th} day in the sample. The notation $\hat{\alpha}, \hat{\beta}$ will be used for the least squares estimates of α and β, with corresponding estimators $\tilde{\alpha}, \tilde{\beta}$.

(a) [3 marks] For each of the following statements, indicate whether they are True, False, or if more information is required (NEI) to determine if they are True or False. You must ensure your answer is clearly marked.

(i) $E[\tilde{\alpha}] = \hat{\alpha}$	True	False	NEI
(ii) $\text{Var}(\tilde{\beta}) \geq 0$	True	False	NEI
(iii) $\tilde{\beta} > \tilde{\alpha}$ if $\overline{x} > \overline{y}$	True	False	NEI
(iv) $P(\tilde{\beta} > \beta) = 0.5$	True	False	NEI
(v) $\tilde{\beta}$ is a continuous random variable	True	False	NEI
(vi) $-2\log\left(\frac{\tilde{\alpha}}{\tilde{\beta}}\right) \sim \chi^2_1$	True	False	NEI

You are given the following R commands and output:

```
> mod <- lm(spend ~ temp)
> summary(mod)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  970.581    104.377   9.299 2.42e-10 ***
temp         -27.807      7.153     (v)  0.00052 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 285.1 on 30 degrees of freedom
Multiple R-squared:  0.335, Adjusted R-squared:  0.3128
```

(b) [5 marks] Based on the preceding output, give the following:

(i) The sample size: _______

(ii) The maximum likelihood estimate of β: _______

(iii) The least squares estimate of β: _______

(iv) s_e (the estimate of σ): _______

(v) The number replaced by `(v)` in the second row of the coefficients table: _______

(vi) The p-value resulting from a test of $H_0 : \beta = 0$: _______

(vii) The probability distribution that is used to calculate the p-value in (vi): _______

(viii) The expected spend on a day where the temperature is 10° Celsius: _____________

For your convenience, some of the preceding output is repeated below:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  970.581    104.377   9.299 2.42e-10 ***
temp         -27.807      7.153     (v)  0.00052 ***
```

(c) [2 marks] Explain, in terms a non-statistician would understand, what the number -27.807 in the output tells us about the relationship between temperature and spend at the store.

(d) [2 marks] Explain, in terms a non-statistician would understand, what the number 970.581 in the output tells us about the relationship between temperature and spend at the store.

Blake proposes fitting a more complex regression model, using the following R command:

```
> mod2 <- lm(spend ~ temp + humidity + sunshine + rain)
```

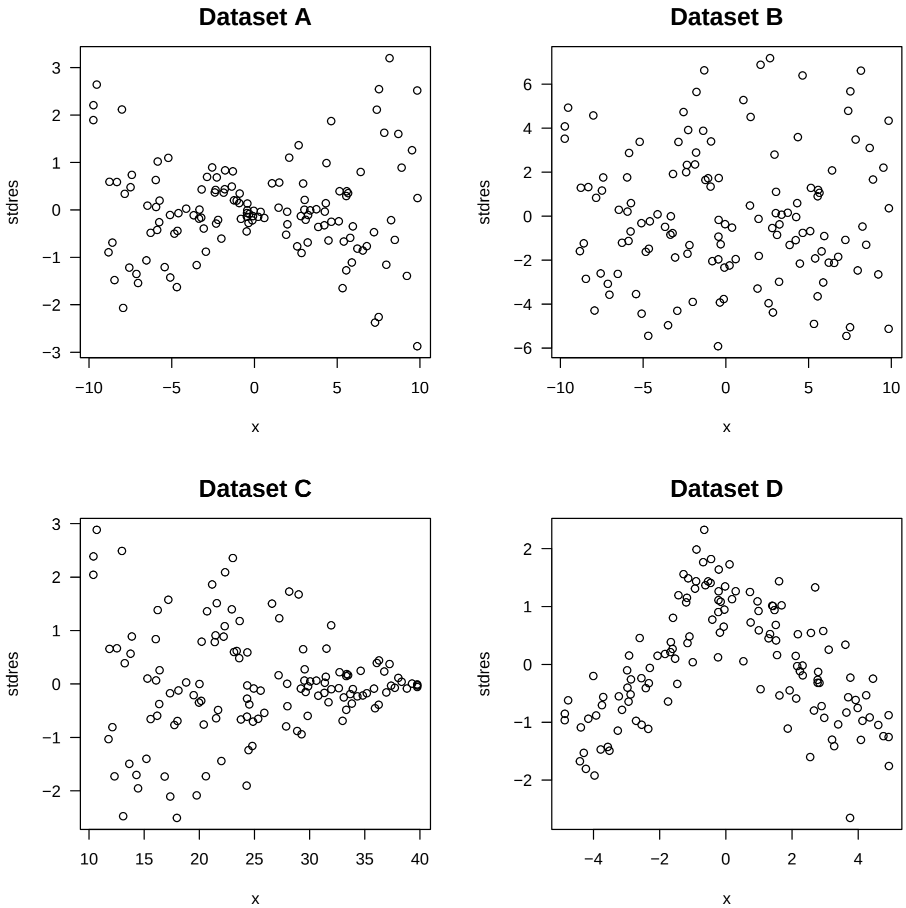
You may assume that `temp`, `humidity`, `sunshine`, and `rain` are all continuous variates stored in R, with the same length as `spend`.

(e) [2 marks] Consider any one of the parameters being estimated by this model. Write down the probability distribution we would use to test the null hypothesis that that parameter's true value is 0.

Note: The final part of this question is unrelated to the analyses carried out in parts (a)-(e).

Four datasets labelled A-D are generated in R, each consisting of 2 variates (`x` and `y`). For each dataset, the following commands are run with `?` replaced with the letter A, B, C, or D, depending on which dataset is being analyzed. Four plots are generated, one for each dataset. These are presented on the following page.

```
mod <- lm(y ~ x)
stdres <- rstandard(mod)
plot(x, stdres, main = "Dataset ?", las = 1)
```



(f) [2 marks] In the table below, insert an X for each dataset where the corresponding assumption appears to be violated based on the above plots, and insert an O if the assumption does not appear to be violated. So, for example, if you thought the linearity assumption was violated for Dataset A, you would insert an X in the first cell in the table. If you thought the linearity assumption was not violated for Dataset A, you would insert an O in the first cell in the table. You must choose X or O for each cell.

Assumption	A	B	C	D
$E[Y_i] = \alpha + \beta x_i$ (linearity)				
$Var(Y_i) = \sigma$ (constant variance)				

Question 7 [16 marks]

Alex and Blake are working on co-op at *Pita Parker*, a new superhero-themed pita restaurant in the UW Plaza. To satisfy a variety of dietary needs, the restaurant offers three types of pita: meat-based, vegetarian, and vegan. Prior to opening, the owners of the restaurant conducted research and estimated that 80% of students eat meat, 15% are vegetarian, and 5% are vegan. The store uses this information to decide how many of each type of ingredient to buy. They will begin by testing the null hypothesis that their sales are consistent with the percentages from their market research. You may assume that each customer only orders 1 pita.

During the first week the store was open, the owners recorded how many of each type of pita was sold, with the following results:

Meat	Vegetarian	Vegan
140	50	10

Let Y_1, Y_2, Y_3 denote the number of times a meat, vegetarian, or vegan pita was ordered, respectively, in a sample of n customers. Let y_j denote the observed value of Y_j. Assume $(Y_1, Y_2, Y_3) \sim \text{Multinomial}(n, \theta_1, \theta_2, \theta_3)$.

(a) [1 mark] Write down the null hypothesis the owners wish to test in terms of the above notation.

Let $\hat{\theta}_j$ denote the maximum likelihood estimate of θ_j based on these data.

(b) [2 marks] Give a numeric value or expression for each of the following, or indicate not enough information (NEI) is provided to give a numeric value or expression.

(i) The maximum likelihood estimate of θ_1:

(ii) $\hat{\theta}_1 + \hat{\theta}_2 + \hat{\theta}_3 =$

(iii) $\theta_1 + \theta_2 + \theta_3 =$

(c) [2 marks] Complete the following table of expected frequencies under the null hypothesis.

	Meat	Vegetarian	Vegan
Observed	140	50	10
Expected			

(d) [4 marks] Carry out a test of the null hypothesis that the advertised probabilities are correct using the Pearson goodness-of-fit test. Your answer should include the following two steps:

(i) A numeric expression for the Pearson goodness-of-fit test statistic:

(ii) A probabilistic expression for the p-value. You may write d for the observed value of the test statistic in (i), if you wish:

You are told that the p-value resulting from the test carried out in part (a) is 0.000365.

(e) [2 marks] You are told that one of the following p-values is the result of testing the same null hypothesis using the likelihood ratio test statistic:

$$p_A = 0.999635 \qquad p_B = 0.365 \qquad p_C = 0.00106$$

Indicate which of these p-values is the one resulting from the likelihood ratio test statistic, and briefly justify your answer.

A few weeks later, the owners ran a promotion to see if it had an effect on the distribution of sales of the three kinds of pita. The promotion offered customers a 50% discount on their next pita purchase if they completed a short survey. The owners did not use the survey data, and instead simply looked at the sales during the week the promotion was running.

(f) [1 mark] Is this an observational study, an experimental study, or a sampe survey? Briefly justify your answer.

The owners observed the following results:

Week	Meat	Vegetarian	Vegan	Total
Opening Week	140	50	10	200
Promotion Week	450	120	30	600

Suppose you wish to test the null hypothesis that the distribution of pita sales is unaffected by the promotion. You may again assume a Multinomial model for these data.

(g) [2 marks] Give numeric expressions or values for the following:

(i) If the null hypothesis is true, the expected number of vegan pitas sold during the opening week:

(ii) If the null hypothesis is true, the expected number of vegetarian pitas sold during the promotion week:

The value of the likelihood ratio test statistic for the test of the null hypothesis with these data is stored in R in an object named `lambda`. You are given the following R commands and output:

```
> pchisq(lambda, 2)
[1] 0.6698439
> dchisq(lambda, 2)
[1] 0.1650781
```

(h) [2 marks] Summarize the conclusion of the hypothesis test. Your answer should clearly indicate the p-value resulting from the test.

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